

the frequency moved up by 1 MHz.

What is important to understand here is: If you're keeping the counter connected to the output of a non-buffered oscillator, the reading of the counter plus $455/10.7$ (as the case may be, FM or AM mode, as shown on the LCD) will be the true reading. Attempting to count the frequency and then disconnecting the counter will set the oscillator at a different frequency after counting due to the capacitance and loading effect of the counter. The reading got then is invalid as the frequency changes on the removal of the counter.

Keeping this effect in mind, it is important to:

1. Use the 2pF capacitor. It causes minimal frequency shift due to minimal capacitance effect, so that the reading obtained, and the frequency after removing the counter from your oscillator, don't differ much.

2. Buffer your oscillator. Buffering of oscillators is very important to prevent your oscillators from being affected by external components.

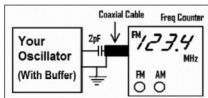


Fig. 13: Suggested connection

Fig. 13 shows the suggested connection. Never connect the HF and VHF counter pins together. Use separate ports and coax cables for both.

Why it's good for beginners?

There are other radio receiver models available with more work to do (in terms of points). Fig. 14 shows an example: As you can see, the number of lines is more, and more connections have to be made in order to get it working. But for a beginner, the KK-939B is a great radio with a great counter board. It has on-board flip-flops that make the external connections simple. In the case of the board shown in Fig. 14, all the flip-flops are on the radio PCB, so the number of connec-



Fig. 14: Another radio receiver example

tions is much more. Besides, separating the counter PCB from the radio PCB removes the luxury of pushbuttons and flip-flops; which are present in the KK-939B.

It took me two month to understand the operation of the counter shown in Fig. 14. After all the UTC6610 learning and circuit decoding (extensive), the KK-939 took only 17 minutes flat. That's why it's good for beginners. Enjoy and have fun with your inexpensive counter. The radio should be available for Rs 250 or less. Have fun! ●

The author is from Manipal Institute of Technology

Statement about ownership and other particulars about

ELECTRONICS FOR YOU

FORM IV (See Rule 8)

- | | |
|--|---|
| 1. Place of publication | : New Delhi |
| 2. Periodicity of its publication | : Monthly |
| 3. Printer's Name | : Ramesh Chopra |
| Nationality | : Indian |
| Address | : ELECTRONICS FOR YOU
D-87/1, Okhla Industrial Area,
Phase I, New Delhi 110020 |
| 4. Publisher's Name | : Same as (3) above |
| Nationality and address | |
| 5. Names and addresses of individuals who own the newspaper & partners or shareholders holding more than 1% of the total capital | : EFY Enterprises Pvt Ltd
D-87/1, Okhla Industrial Area,
Phase 1, New Delhi 110020 |

I, Ramesh Chopra, hereby declare that the particulars given above are true to the best of my knowledge and belief.

Date: 28-2-2009

Ramesh Chopra
Publisher